Appendix · pumllint for the PoC · what we decide

# A PoC on semantic linting: go or no-go?

## WHY, IN THREE LINES

| | |
|---|---|
| 1 **The problem** | Diagrams render, but not all make sense; every reader downstream, human or AI, fills the gaps. |
| 2 **The contribution** | A semantic linter finds model smells, gives each diagram a maturity level and gates the hand-off, in the editor and in CI. |
| 3 **Measured** | AI code from Level 1 sequence diagrams fails ~1 in 3 intended behaviours; from Level 2 up, ~1 in 10. |

## YOUR TEAM

| | |
|---|---|
| **Team / squad** | … |
| **Team contact** | *name, role* |
| **Raised by** | *round-table attendee* |
| **Date** | … |

**Your answer** ☐ **Go** ☐ **No-go** ☐ **Not yet** **What would change our mind?** …

## IF GO, WE AGREE ON

| Scope | Success criteria | Owner & timebox | Next check-in |
|---|---|---|---|
| *which team, which diagrams* | *what must the PoC show?* | *who leads, how many weeks* | *when you look at the results* |

Return to: **[presenter name]** · **[email]** · by **[reply-by date]**

*Getting started: the three pumllint slides that follow*

Appendix · pumllint for the PoC · install and first run

# Get started in ten minutes

## SIX COMMANDS

```
$ pip install pumllint            # or pipx / uv tool install
$ pumllint diagrams/              # findings, with severity
$ pumllint score diagrams/        # maturity level per diagram
$ pumllint score diagrams/ -f html -o report.html
$ pumllint --list-rules           # everything it checks
$ pumllint lsp                    # findings in your editor
```

## WHAT YOU GET BACK

**Findings**
rule, severity and line, e.g. ACT002 · major

**A maturity level**
per diagram and per set, with what it takes to reach the next one

**An HTML report**
one self-contained page, to share or attach to a review

**Exit codes for CI**
0 clean · 1 findings at or above --fail-on (default major) · 2 usage error

**Python 3.11 or later · no runtime dependencies (PyYAML only for a YAML config).** Read more in the repository: README · docs/findings-and-scores.md · docs/rule-map.md · docs/business-processes.md

# Fit it to your team

## Configure

pumllint.yaml (or .toml / .json) is picked up automatically:

```
rules:
  unnamed-diagram: false
  participant-naming:
    severity: major
  requirement-link:
    pattern: "REQ-\d+"
```

**One exception in one diagram:** `' pumllint: disable=SEQ006`

## Automate

GitHub Actions, pinned to a release:

```
- uses: fdurieux/pumllint@v0.35.0
  with:
    command: score
    paths: docs/diagrams
    baseline: maturity.json
    min-level: "2"
```

**Or per commit:** the pre-commit hooks pumllint and pumllint-score.

## Connect

**SonarQube**
-f sonar writes SonarQube's import format: findings on the same dashboard and in pull requests.

**ARIS**
the separate aris2puml tool converts processes to PlantUML; they pass the same gate.

**Editors**
pumllint lsp in any editor that supports LSP; no editor-specific plugins.

**Step 1 · ADVISE** report only, no gate → **Step 2 · RATCHET** `--baseline maturity.json` → **Step 3 · FLOOR** `--min-level 2`

Appendix · pumllint for the PoC · a suggested plan

# A four-week PoC, and what to measure

| Week 1 · Baseline | Week 2 · Advise | Week 3 · Ratchet | Week 4 · Review |
|---|---|---|---|
| One team and its diagrams. Score them, publish the HTML report, record maturity.json. | pumllint lsp in the editors; authors fix as they go. Nothing fails. | In CI with --baseline: no diagram may get worse. | Compare with the baseline; agree the floor and the next step. |

## What to measure

- Findings per dimension, week 1 against week 4
- Sequence and state diagrams at Level 4; processes without a major finding
- Time to fix a finding, as the authors report it
- Noise: findings the team disagrees with, and the config changes they lead to
- Feedback from authors and reviewers
- Rework found downstream: the expected benefit, measured here for the first time

**Success criteria**
Agreed on the closing slide: what must the PoC show? Write them here before week 1, so week 4 compares against them.

**A proposal to adapt:** the length, the team and the measures are the PoC owner's call.